A Physics-Consistent Machine Learning Framework for High-Resolution Ozone Downscaling in the Greater Bay Area

Wentao Wang*
Equal Contribution

Lifeng Han*
Equal Contribution

Chenxi Pan*
Equal Contribution

***Abstract*—This study aims to address the limitations of current chemical transport models (CTMs) in ozone forecasting, including computational time-consuming nature, low spatial resolution, and insufficient long-term prediction accuracy. Whilst traditional CTMs hold value for early warnings, their operational constraints necessitate the development of more efficient hybrid forecasting frameworks. Given ozone's severe threat to public respiratory health, particularly against the backdrop of persistently rising ozone levels in Hong Kong and the Guangdong-Hong Kong-Macao Greater Bay Area (GBA), enhancing forecast accuracy is imperative.**

This study proposes a machine learning-based high-precision downscaling framework that deeply integrates GEOS-Chem simulation outputs with high-resolution meteorological parameters and ground-level pollutant observations. We designed systematic model comparison experiments encompassing ensemble tree models (represented by Random Forest and XGBoost) and neural network architectures (represented by MLP and DNN). Experiments were conducted in two scenarios: locally in Hong Kong (data-constrained) and across the entire Greater Bay Area (data-rich), aiming to evaluate model robustness and generalisation capability, supplemented by rigorous hyperparameter optimisation and Model Confidence Set (MCS) validation.

Results demonstrate that the XGBoost model exhibits superior robustness and generalisation across all scenarios. Notably, in the Greater Bay Area-wide test, this ensemble model achieves remarkable predictive accuracy ($R^2 \approx 0.86$), substantially outperforming traditional GEOS-Chem simulations ($R^2 \approx 0.29$). Moreover, ensemble tree models consistently outperformed neural networks, a finding consistent with existing literature and demonstrating their advantage in processing structured environmental data.

This study employs SHAP-based interpretability analysis to address the black-box issue in machine learning, demonstrating that the model captures physically meaningful patterns consistent with atmospheric chemistry: temperature was identified as the primary driver of ozone formation, while nitrogen dioxide (NO_2) exhibited a pronounced non-linear titration effect under locally high concentrations. This finding corrects previous research's exclusive emphasis on emission sources, demonstrating the potential of data-driven models in understanding complex physicochemical coupling mechanisms.

This study not only provides an operational high-resolution ozone forecasting tool but also offers a robust theoretical foundation for scientific decision-making in regional air quality management.

***Index Terms*—ozone, GEOS-Chem, machine learning, XGBoost, downscaling, SHAP**

I. Introduction

A. Background and Motivation

In recent decades, a growing body of scientific evidence has indicated that surface ozone pollution has become an increasingly severe environmental issue. While concentrations of certain pollutants such as $PM_{2.5}$ and CO have shown signs of decline due to stringent control measures, surface ozone levels have continued to rise. This trend poses a significant threat to both environmental quality and public health. Exposure to elevated ozone concentrations has been linked to respiratory diseases, cardiovascular issues, and increased mortality, particularly among vulnerable populations such as children and the elderly. As a result, effective monitoring, modeling, and prediction of surface ozone concentrations have become crucial for developing robust mitigation strategies.

Numerous studies have identified anthropogenic activity as a key driver of increased ground-level ozone concentrations. For instance, the imbalance between nitrogen oxides (NO_x) and volatile organic compounds (VOCs) in emission control strategies can lead to unintended secondary ozone formation (Wang et al., 2019). Additionally, urbanization contributes to the heat island effect and elevates surface temperatures, further intensifying ozone production in regions with dense traffic and industrial activities. These human-induced changes interact with natural atmospheric processes in complex ways, making it increasingly difficult to predict ozone behavior using traditional empirical methods.

Meteorological factors also play a critical role in ozone dynamics. Tu et al. (2007) found that temperature and wind speed are significantly positively correlated with ozone levels, while relative humidity and precipitation generally exhibit negative correlations. However, seasonal differences exist: in winter, weak photochemical activity reduces the significance of these correlations, and ozone variation becomes more influenced by boundary-layer dynamics and human emissions. This seasonal and spatial variability highlights the need for models that can capture both chemical and physical interactions within the atmosphere.

Traditionally, chemical transport models (CTMs) such as GEOS-Chem have been employed to simulate atmospheric ozone distribution. These models numerically solve advection-

diffusion-reaction equations to describe the physical and chemical transformation of pollutants over time and space (Watson-Parris et al., 2021). CTMs are grounded in fundamental atmospheric chemistry and physics, making them highly interpretable and comprehensive. However, they face several limitations. High computational demand restricts their use in high-resolution or long-term applications (Lyu et al., 2024). Furthermore, uncertainties in input datasets—such as emissions inventories and meteorological forcings—can lead to substantial deviations from observed values (Huang et al., 2024). Moreover, the grid resolution of global CTMs is typically around 100 km, which limits their ability to resolve localized air pollution hotspots, especially in urban areas (Ryan & Wild, 2021).

B. Research Objectives

Given the limitations of traditional CTMs, this study aims to develop a hybrid modelling framework capable of significantly enhancing predictive resolution, efficiency, and accuracy. Unlike previous applied research focused solely on single algorithms, this study sets three core objectives:

Constructing a multi-model arena and benchmarking
Introducing ensemble tree models represented by Random Forest and XGBoost, and systematically comparing them with neural network architectures represented by Multi-Layer Perceptrons (MLP) and Deep Neural Networks (DNN). This aims to investigate performance differences and applicability boundaries across distinct algorithms when processing structured atmospheric environmental data.

Validating spatial generalisation capability and statistical significance
Designing two experimental scenarios — Hong Kong local (data-constrained) and Greater Bay Area-wide (data-rich) — to validate the critical role of spatial information expansion in enhancing prediction accuracy. Concurrently, introducing Model Confidence Sets (MCS) testing to rigorously identify optimal models from a statistical perspective, thereby avoiding misjudgements stemming from reliance on point estimation metrics alone.

Model Interpretability and Physical Consistency
To address the interpretability limitations commonly associated with machine learning models, SHAP (SHapley Additive exPlanations) was employed to analyse the internal decision mechanisms of the selected model. This analysis aimed to assess whether the learned relationships are physically consistent with established atmospheric chemistry principles, including temperature-dependent photochemical reactions and the nonlinear titration effect of NO_2, rather than representing purely statistical associations.

C. Structure of the Report

This report is organized into seven main sections. The **Introduction** outlines the background, motivation, and objectives of the study. The **Literature Review** summarizes recent research on ozone prediction models, downscaling methods, and relevant variables. The **Methodology** section details the data sources, preprocessing steps, model design, and evaluation framework. The **Results and Discussion** present model performance, comparison with GEOS-Chem outputs, and key insights. The **Conclusion** summarizes the findings, addresses challenges, and proposes future work. Supplementary details, including figures and codes, are included in the **Appendix**.

II. Literature Review

For this section, the literature review is organised into three components. First, studies on machine learning algorithms are synthesised to clarify how existing modelling strategies inform the selection of predictive methods in this research. Second, methodological developments in downscaling and model–data fusion are examined to highlight the frameworks that guide the design of our hybrid modelling approach. Finally, prior findings on meteorological and chemical drivers of ozone formation are reviewed to justify the feature set adopted in the predictive model.

A. AI and Algorithm

Recent years have witnessed growing interest in the application of machine learning (ML) techniques to enhance air quality prediction, particularly for surface ozone (O_3), whose formation is driven by complex non-linear interactions among meteorological conditions and precursor pollutants. Unlike traditional statistical models that assume linearity, ML algorithms such as support vector regression (SVR), random forests (RF), and gradient boosting machines (GBM) are capable of modeling high-dimensional, non-linear relationships with greater flexibility. In particular, gradient boosting methods—such as LightGBM and CatBoost—have been increasingly adopted due to their ability to handle heterogeneous data, missing values, and categorical features without extensive preprocessing.

One critical determinant of model performance lies in the choice and tuning of hyperparameters. Studies have shown that the effectiveness of boosting-based models is highly sensitive to parameter choices including tree depth, learning rate, regularization terms, and number of estimators. As a result, hyperparameter optimization is not merely an auxiliary step but a core component of the modeling process. Recent research has employed Bayesian optimization and evolutionary algorithms to automatically explore hyperparameter spaces, enabling more robust model generalization across spatial and temporal domains. For instance, Zhang et al. (2022) demonstrated that a LightGBM model with Bayesian-tuned parameters significantly outperformed manually tuned models in daily ozone prediction across eastern China. Similarly, a study by Fang et al. (2023) applied a genetic algorithm for hyperparameter selection in an ensemble learning model and reported enhanced forecasting accuracy under varying meteorological regimes.

These developments are particularly relevant to the present study, which employs the XGBoost algorithm to downscale GEOS-Chem ozone simulations in the Greater Bay Area of

China. Given the region's pronounced meteorological variability and urban–rural emission gradients, the success of the hybrid model hinges not only on variable selection but also on the effective tuning of model parameters. Incorporating an automated or grid-based optimization strategy ensures that the boosting model adapts to local nonlinearities while preserving generalizability. Therefore, aligning the ML configuration with physical model outputs represents a key step toward achieving both high-resolution and high-fidelity ozone forecasts in complex urban environments.

Inspired by Zhang et al. (2022) and Fang et al. (2023), this study adopted a structured hyperparameter tuning approach using randomized search with cross-validation. Rather than relying on default or manually assigned parameters, key boosting-related parameters—such as tree depth, learning rate, and regularization coefficients—were optimized to enhance generalization and stability across the diverse spatiotemporal patterns of ozone concentrations in the Greater Bay Area.

B. Methodology

The integration of machine learning into atmospheric modeling has led to significant advances in the accuracy and resolution of surface ozone (O_3) predictions. Gouldsbrough et al. (2024) developed a robust methodological framework that downscales the EMEP4UK chemical transport model (CTM) output from a coarse 5×5 km grid to a finer 1×1 km resolution using gradient-boosted trees (GBTs). This approach effectively corrected for the positive ozone bias in CTM predictions, especially in urban areas, by combining CTM data with observational measurements and meteorological covariates.

A notable advantage of GBTs, particularly the XGBoost implementation, is their capacity to capture complex nonlinear relationships across high-dimensional inputs while maintaining computational efficiency (Chen & Guestrin, 2016). To tailor model performance, the authors adopted a two-stage hyperparameter tuning strategy: Bayesian optimization via Hyperopt followed by manual fine-tuning. Additionally, they used gamma regression to address the long-tailed distribution of ozone data and applied data resampling to mitigate bias at extreme ozone concentrations.

The methodology involved spatial interpolation of CTM outputs, alignment of grid cells with monitoring stations, and the selection of relevant features such as 2m maximum temperature, wind components, PBL height, and land-use proximity metrics. The final model demonstrated a substantial increase in predictive R^2 and a reduction in RMSE compared to the original CTM simulation, thereby establishing a scalable post-processing tool for ozone surface generation.

Xiong et al. (2024) further supported this methodological direction by comparing LightGBM, XGBoost, and Random Forest algorithms for ozone prediction in China. Their results revealed that LightGBM offered superior computational efficiency and prediction accuracy when downscaling CMAQ model outputs, achieving R values up to 0.88 for daily maximum 8-hour ozone (O_3-Max8h). Importantly, their variable selection emphasized avoiding multicollinearity among pollutants, with features such as NO_2, CO, and radiation being prioritized due to their known mechanistic relevance to ozone formation.

Another study by Liu et al. (2022) illustrated a nationwide downscaling strategy using XGBoost to generate $0.1^\circ \times 0.1^\circ$ resolution ozone data in China, incorporating both spatiotemporal terms and observational measurements. This reinforced the efficacy of ensemble-based machine learning models in capturing the dynamic behavior of O_3, especially when used to complement or correct process-based CTM outputs.

These methodologies directly informed the design of this thesis. For example, we adopted GBTs (XGBoost) as the predictive algorithm, carefully tuned hyperparameters via cross-validation, and excluded time-independent variables from scaling. Moreover, our feature set was aligned with those identified as most impactful by prior SHAP analyses, such as temperature and boundary layer height, to ensure model interpretability and physical plausibility. This strategy aims to enhance the resolution and accuracy of ozone prediction over the Greater Bay Area, where traditional CTMs are limited by coarse spatial grids and bias from insufficient NO_x titration.

C. Meteorological Variables

Meteorological conditions play a critical role in shaping surface ozone (O_3) concentrations, making them indispensable variables in any modeling framework aiming to accurately predict ozone dynamics. Among these, temperature, relative humidity, wind speed, and solar radiation are particularly influential due to their impact on photochemical reactions and pollutant dispersion.

Temperature has a well-documented positive correlation with ozone concentrations, particularly during the warmer months. As documented by Tu et al. (2007), surface ozone levels in Nanjing increased significantly with rising temperature, attributed to intensified photochemical reactions. The study also noted that temperatures above 30°C yielded the highest O_3 levels, while lower temperatures were associated with elevated concentrations of ozone precursors like NO and NO_2. This thermal dependency is also reflected in the enhanced decomposition of peroxyacetyl nitrate (PAN), a key ozone precursor reservoir, under high-temperature conditions, further amplifying O_3 production in polluted regions.

Relative humidity, on the other hand, demonstrates a negative relationship with ozone concentrations. Low humidity is often linked with higher O_3 levels due to reduced wet deposition and a more efficient ultraviolet actinic flux, which drives photochemical ozone production. Empirical evidence from Tu et al. (2007) revealed that O_3 concentrations decreased from 43.4 ppbv at $\leq$40% humidity to just 10.9 ppbv at $>$80% humidity, reinforcing the significance of this inverse relationship.

Wind speed is another crucial variable, acting both as a facilitator of pollutant dispersion and a contributor to vertical mixing. Moderate wind speeds (2–4 m/s) have been shown to elevate surface ozone levels by enhancing mixing between

atmospheric layers, whereas very high speeds tend to dilute concentrations through increased dispersion.

Moreover, solar radiation and boundary layer height, although not always explicitly measured, are indirectly represented through temperature and wind variables. These influence the rate of photolysis and the depth of pollutant mixing, making their proxies essential in data-driven models.

The inclusion of meteorological features in your XGBoost-based ozone prediction model is therefore strongly justified, not only due to their direct statistical correlations but also because of their mechanistic ties to ozone photochemistry and transport dynamics. Their integration enhances both the physical interpretability and predictive accuracy of machine learning models, especially when correcting biases inherent in chemical transport model outputs.

III. Methodology, Data Collection and Analyses

This study aims to improve the prediction of surface ozone (O_3) concentrations over the Greater Bay Area by integrating meteorological and pollutant observations into a machine learning framework, using GEOS-Chem simulation results as a base layer. The methodology consists of three key phases: data acquisition and preprocessing, model construction, and predictive evaluation.

A. Data Sources and Features

Meteorological data were obtained from the ERA5 reanalysis dataset, provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). We selected five key meteorological parameters known to influence ozone formation and transport: 2-meter air temperature (t2m), surface pressure (SP), relative humidity (r), 10-meter wind components (U10M and V10M), and ultraviolet radiation (UVB). These variables were extracted at hourly resolution for January 2020 and subsequently averaged on a daily basis to reduce data volume and suppress short-term noise.

In addition to meteorological inputs, surface-level pollutant observations were collected from regional monitoring stations within the Greater Bay Area. The pollutant variables included nitrogen dioxide (NO_2), sulfur dioxide (SO_2), fine particulate matter ($PM_{2.5}$), and ozone (O_3). Among these, only ozone served as the prediction target, while the remaining variables were used as predictive features during model training.

B. Downscaling and Data Harmonization

The GEOS-Chem model output, which provided baseline ozone concentrations, was originally generated at a spatial resolution of $0.25^\circ \times 0.25^\circ$. Similarly, ERA5 meteorological fields were provided at 0.25° resolution but resampled to match the coarser grid. To ensure spatial comparability across all data sources, we applied bilinear interpolation to uniformly downscale all gridded variables to a $0.01^\circ \times 0.01^\circ$ resolution. This resolution was chosen as a trade-off between computational efficiency and spatial accuracy, especially for regional-scale modeling over southern China.

Following spatial harmonization, all datasets were aligned temporally using date indices. Daily mean values were calculated from hourly records for each variable, both to facilitate model convergence and to match the resolution of the observational data.

C. Model Integration and Justification

The core experimental design of this study is established upon a **multi-dimensional model comparison framework**. To investigate the advantages of different algorithmic architectures, we selected four mainstream machine learning models: **Random Forest** as the baseline, **XGBoost** as the representative of advanced gradient boosting, and **MLP** and **DNN** representing the neural network direction.

To ensure a fair comparison, all models were placed within a unified **StandardScaler** pipeline. We employed **3-fold cross-validation** combined with the **RandomisedSearchCV** strategy for hyperparameter tuning. Specifically, the training data was randomly partitioned into three subsets. Models were trained on two subsets and validated on the remaining subset, with this process repeated to assess parameter robustness.

D. Model Integration, Standardization, and Justification

The predictive model was built using the XGBoost algorithm, which was trained on a feature set comprising downscaled meteorological and pollutant data. The target variable was the observed daily ozone concentration from ground stations. To improve model convergence and ensure numerical stability, all continuous variables, including meteorological factors and pollutant concentrations, were standardized using z-score normalization, defined as:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where x is the raw value, μ is the mean, and σ is the standard deviation computed from the training set.

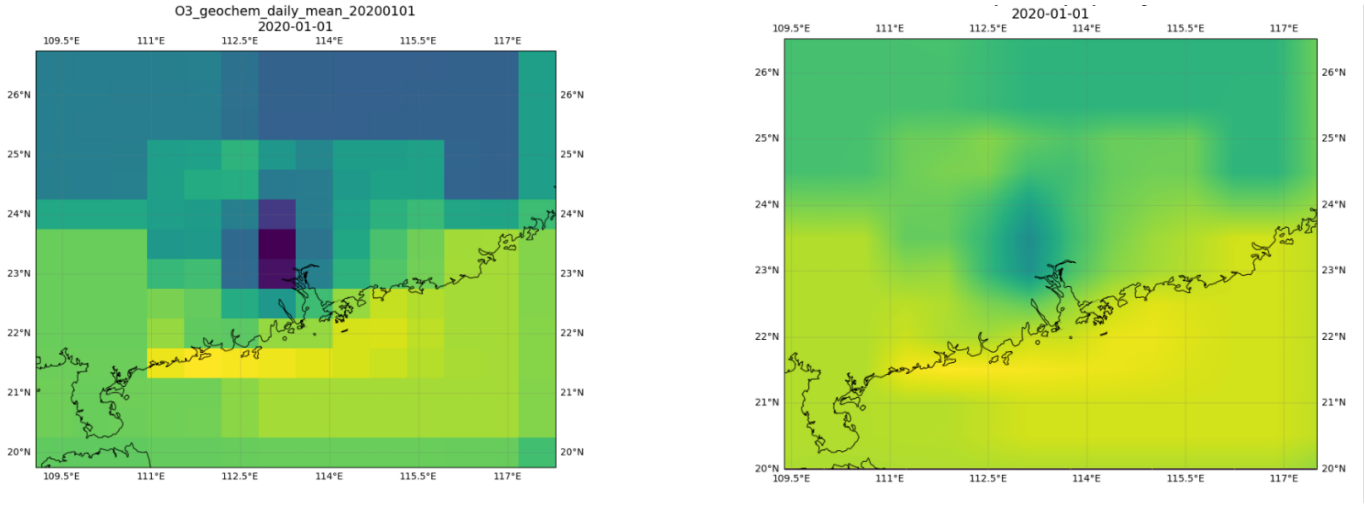
The only variables excluded from standardization were categorical or temporally static features, such as the year, month, day, and spatial coordinates (latitude and longitude), to preserve interpretability. Standardization was implemented using `sklearn.preprocessing.StandardScaler()` within the modeling pipeline, prior to cross-validation.

E. SHAP-Based Model Interpretation

To improve model transparency and identify the most influential variables, we performed a post-hoc SHAP (SHapley Additive exPlanations) analysis. Using the **TreeExplainer**, we computed the contribution of each feature to the model's prediction across the test set. Bar and summary plots were generated to visualize global importance and feature interaction effects. SHAP values confirmed that temperature, NO_2, and GEOS-Chem baseline ozone were consistently among the top contributors to model predictions.

TABLE I: **Features and Data Source**

Input Features	Source	Abbreviation	Resolution
Longitude	China-site	Lon	–
Latitude	China-site	Lat	–
Daily mean SO_2 observation concentration	China-site	SO2_obs	–
Daily mean $PM_{2.5}$ observation concentration	China-site	PM2.5_obs	–
Daily mean NO_2 observation concentration	China-site	NO2_obs	–
Daily mean temperature in 2m	ERA5	t2m	$0.01° \times 0.01°$
Daily mean wind speed (x component)	ERA5	U10M	$0.01° \times 0.01°$
Daily mean wind speed (y component)	ERA5	V10M	$0.01° \times 0.01°$
Daily mean pressure	ERA5	SP	$0.01° \times 0.01°$
Daily mean 1000hPa humidity	ERA5	r	$0.01° \times 0.01°$
Daily mean UV radiation	ERA5	uvb	$0.01° \times 0.01°$
Daily mean surface O_3 in GEO-chem	GEO-chem	O3_gc	$0.01° \times 0.01°$
Year	GEO-chem	year	–
Month	GEO-chem	month	–
Day	GEO-chem	day	–



(a) O_3 GEOS-Chem, 2020-01-01

(b) Downscaled O_3, 2020-01-01

Fig. 1: **Regridded comparison**

F. Model Evaluation and Error Analysis

This study primarily employs the **Coefficient of Determination (R^2)** as the core metric for assessing model predictive accuracy, serving to quantify the model's explanatory power over the variance in observed data.

Given that variations in R^2 across different models may stem from randomness, a mere comparison of numerical values is insufficient for robust conclusions. Consequently, this study further introduces the **Model Confidence Set (MCS)** as a statistical testing tool. The MCS effectively mitigates misjudgements of model performance arising from random sample fluctuations.

At a given confidence level (set at $\alpha = 0.1$, i.e., 90% confidence in this study), the MCS constructs a set encompassing all "statistically indistinguishable optimal models" based on the prediction loss function (squared error). If a model's p-value is less than 0.1, it is statistically significantly inferior to the optimal model and is thus excluded; conversely, if the p-value exceeds 0.1, the model is retained within the confidence set.

This methodology provides robust statistical evidence for determining the final best model.

IV. Procedures

Building on the methodological design outlined in the previous section, this section details the data-processing and modelling steps used to construct the final dataset, train the predictive models, and generate all diagnostic figures. The workflow follows a sequential structure: (i) preprocessing of GEOS-Chem and ERA5 fields, (ii) temporal and spatial matching with ground-based observations, (iii) feature engineering consistent with Table I, (iv) model training and hyperparameter

optimisation under two experimental scenarios, and (v) post-hoc statistical and interpretability analyses.

A. Preprocessing of GEOS-Chem and ERA5 Fields

In the first stage, raw GEOS-Chem outputs and ERA5 reanalysis fields were converted into daily products suitable for subsequent matching and downscaling.

For GEOS-Chem, three-dimensional ozone mass mixing ratios were extracted from the original NetCDF files. After harmonising the coordinate system to a common longitude–latitude–time convention, the ozone field was converted from mass mixing ratio ($\mathrm{kg\,kg^{-1}}$) to near-surface concentration ($\mu\mathrm{g\,m^{-3}}$) using a constant air density of $1.18\,\mathrm{kg\,m^{-3}}$:

$$\mathrm{O_3\ (\mu g\ m^{-3})} = \mathrm{O_3\ (kg\ kg^{-1})} \times \rho_{\mathrm{air}} \times 10^9.$$

The resulting near-surface ozone concentrations were stored as daily fields on the GEOS-Chem grid and, together with the meteorological variables described below, were subsequently interpolated consistently with the $0.01^\circ \times 0.01^\circ$ analysis grid adopted in Table I and Section III-B.

For meteorology, ERA5 data (including 2 m air temperature, 10 m wind components, surface pressure and auxiliary fields) were imported at hourly resolution and reprojected onto a consistent longitude–latitude–time framework. Daily mean fields were then computed by resampling along the time dimension with a 1-day window and taking the arithmetic mean. The resulting daily meteorological dataset was aligned temporally with the GEOS-Chem ozone product and regridded using bilinear interpolation so that all gridded predictors share a common spatial support over the Greater Bay Area. When humidity and radiation datasets were available, they were processed using the same daily-mean resampling and regridding procedure.

B. Spatio-temporal Matching with Ground Observations

In the second stage, ground-based pollutant measurements were matched to the gridded model fields to form a unified station-level dataset. Hourly or daily observations of ozone, nitrogen dioxide, sulphur dioxide and fine particulate matter ($PM_{2.5}$) from monitoring stations across the Greater Bay Area were first ingested from a master observational database. Station identifiers, geographical coordinates and time stamps were checked for consistency, and observation dates were converted into a standard calendar-date variable.

For each station–day record, the geographic coordinates and date were used to obtain collocated GEOS-Chem and ERA5 values. To accommodate model grids stored on a 0–360° longitudinal system, negative station longitudes were shifted by 360° whenever required. Spatial interpolation was performed bilinearly in longitude–latitude space on the common analysis grid, while nearest-neighbour selection was used along the daily time axis to account for minor discrepancies in time stamps.

Each matched entry therefore contains station coordinates, observed ozone as the target variable, co-pollutants (NO_2, SO_2, $PM_{2.5}$), GEOS-Chem baseline ozone, and a full set of collocated meteorological variables (including temperature, wind components, surface pressure, relative humidity and ultraviolet radiation). Records with missing values in key fields, particularly ozone or NO_2, were removed. The resulting station–day table served as the basis for subsequent feature engineering.

C. Feature Engineering and Outlier Treatment

Feature engineering was designed to align strictly with the feature set listed in Table I. First, temporal attributes including calendar year, month and day were derived from the observation date and appended as integer-valued predictors. Second, meteorological variables were harmonised to a unified naming convention (for example, distinguishing zonal and meridional 10 m winds, and assigning a dedicated variable for ultraviolet radiation), such that downstream analyses and interpretability diagnostics use consistent labels across figures and tables.

A strict feature whitelist was then applied to retain only those variables used in the final model configuration: longitude and latitude, co-pollutants (NO_2, SO_2, $PM_{2.5}$), near-surface temperature, 10 m wind components, surface pressure, relative humidity, ultraviolet radiation, GEOS-Chem baseline ozone, and the three calendar variables (year, month and day). Observed ozone remained the sole prediction target. Before model training, a univariate outlier filter was applied to ozone: station–day samples with absolute z-scores larger than 3 were discarded. This step reduced the influence of extreme observations while preserving the overall structure of the ozone distribution.

D. Experimental Scenarios and Train–Test Splitting

To evaluate both local and regional generalisation, two experimental scenarios were defined based on the matched dataset. In the Hong Kong scenario, a geographic filter was applied to retain only stations located within a predefined bounding box surrounding Hong Kong (22.15°–22.57°N, 113.80°–114.50°E). Provided that at least 100 valid station–day samples were available, this subset was used for model comparison experiments.

In the Greater Bay Area scenario, all valid samples from the matched dataset were included without additional spatial restriction. For both scenarios, the data were randomly shuffled using a fixed random seed and then split into training (80%) and test (20%) sets. The same random seed and splitting procedure were used across all models so that any observed performance differences arise from the modelling choices rather than from variability in the train–test partition.

In addition to the multi-model comparison experiments, a dedicated bias-correction configuration was considered for XGBoost. In this configuration, only the Greater Bay Area scenario was used, and the same 80/20 train–test split as above was retained; the difference lies in the more detailed hyperparameter search and time-series validation strategy described in the next subsection. The resulting fitted model was used to

generate the key figures and statistics reported in the main text and appendix.

E. Model Training and Hyperparameter Optimisation

Model training proceeded in two stages corresponding to (i) multi-model benchmarking and (ii) the construction of a single operational bias-correction model based on XGBoost.

For the benchmarking stage, four candidate algorithms were considered under each experimental scenario: random forest, XGBoost, a shallow multilayer perceptron (MLP) and a deeper feed-forward neural network (DNN). For each algorithm, all continuous predictors were standardised to zero mean and unit variance based on the training set before fitting. Hyperparameters were tuned using randomised search with three-fold cross-validation and the coefficient of determination (R^2) as the optimisation criterion. For each learner, eight hyperparameter combinations were sampled from predefined search spaces: tree depth and learning rate for the gradient-boosted trees, number of trees and maximum depth for the random forest, and hidden-layer sizes, regularisation strength and learning rate for the MLP and DNN. The best configuration for each model, as measured by cross-validated R^2, was retained and evaluated on the held-out test set.

In the bias-correction stage, attention was restricted to XGBoost, which was refitted using the Greater Bay Area training set. The same standardisation of continuous predictors was applied. Hyperparameters were selected through a more extensive randomised search combined with a time-series cross-validation scheme that respects the chronological structure of the data. Specifically, a three-fold temporal split was used to avoid information leakage from future observations into the training period. The search explored an enlarged hyperparameter grid, including the number of boosting iterations, learning rate, maximum tree depth, subsampling ratios and column subsampling, again using R^2 as the scoring metric. The best-performing configuration was saved and used as the reference model for subsequent evaluation and interpretation.

F. Performance Assessment, MCS Testing and SHAP Analysis

Model performance in both scenarios was first quantified on the test set using R^2, mean absolute error (MAE) and root mean squared error (RMSE). For each model and scenario, prediction errors were additionally summarised using residual histograms and observed–predicted scatter plots with 1:1 reference lines. These diagnostics served to assess overall bias, dispersion and potential heteroscedasticity.

To formally compare models, the Model Confidence Set (MCS) procedure was applied to the squared prediction errors on the test set. For each scenario, the loss series from all candidate models were assembled into a panel, and the MCS algorithm was run at a significance level of $\alpha = 0.1$. The resulting p-values and inclusion flags were tabulated, allowing us to distinguish models that are statistically indistinguishable from the best-performing specification from those that are significantly inferior. This procedure underpins the model ranking reported in Table II and guards against overinterpreting small differences in point estimates of R^2.

Finally, to mitigate the typical black-box character of machine learning models and to link predictive performance back to atmospheric process understanding, SHAP-based interpretability analysis was carried out on the best-performing XGBoost configuration trained under the Greater Bay Area scenario. The internal tree-based regressor and the associated standardisation step were extracted, and SHAP values were computed for all features in the test set. Global explanations were obtained through SHAP summary plots and bar plots of feature importance, while local mechanisms were examined using waterfall plots for representative high-ozone cases. Additional SHAP dependence plots were produced for key variables such as temperature, ultraviolet radiation and NO_2, including pairwise interaction plots. These outputs form the basis of the physical interpretability analysis and black-box diagnostics discussed in the Results and Discussion section.

V. Result and discussion

A. Model Benchmarking and Performance Evaluation

To quantitatively compare the four candidate models under the two experimental scenarios introduced in Section III, we first summarise their out-of-sample performance using the Model Confidence Set (MCS) framework. Table II reports, for each scenario–model combination, the test-set R^2, the MCS p-value based on squared prediction errors, and an indicator of whether the specification is retained in the 90% confidence set.

TABLE II: Model Confidence Set (MCS) Analysis

Scenario	Model	R^2	MCS p-value	In 90% MCS
GBA_Scenario	XGBoost	0.858	[1.]	[True]
GBA_Scenario	RandomForest	0.831	[0.036]	[False]
GBA_Scenario	MLP	0.792	[0.009]	[False]
GBA_Scenario	DNN	0.792	[0.009]	[False]
HK_Scenario	XGBoost	0.802	[1.]	[True]
HK_Scenario	RandomForest	0.801	[0.964]	[True]
HK_Scenario	DNN	0.785	[0.953]	[True]
HK_Scenario	MLP	0.593	[0.017]	[False]

The experimental results clearly demonstrate that the expansion of spatial scope exerts a decisive influence on model performance, and, at the same time, sharpens the contrast between different algorithmic families. This is consistent with the methodological design in Section III, where the Hong Kong and Greater Bay Area (GBA) scenarios were constructed to probe the role of data volume and spatial heterogeneity in shaping predictive skill.

Within the limited dataset of the local Hong Kong scenario, the models exhibit relatively balanced performance, forming a near three-way equilibrium among the best-performing candidates. Specifically, the R^2 values for DNN, random forests, and gradient boosting trees (XGBoost) on the test set all fall between 0.785 and 0.802, with the MLP lagging

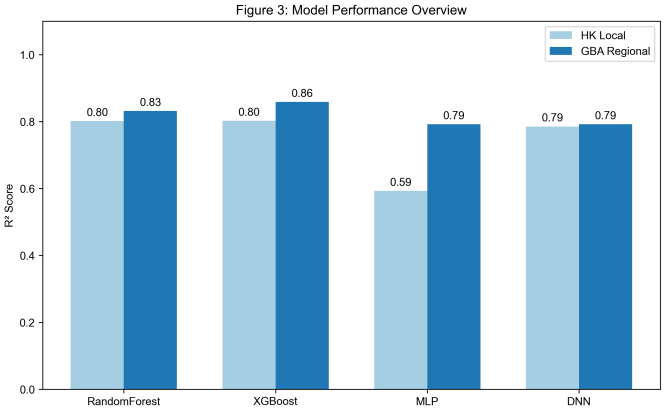


Fig. 2: Model Benchmarking Bar

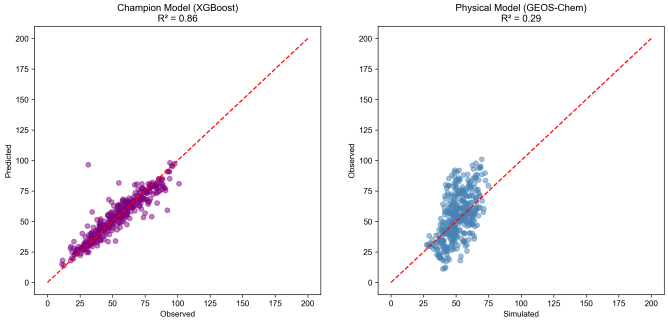


Fig. 3: Champion vs Physics

behind at 0.593 (Table II). Based on the multiple comparison significance (MCS) testing, the p-values for XGBoost, random forest, and DNN all exceed 0.1, indicating no statistically significant difference in predictive performance among these three models under small-sample conditions. In other words, once sampling variability is accounted for, they form a statistically indistinguishable group. Notably, neural network models do not demonstrate the anticipated advantage in this data-constrained regime: the MLP is excluded from the 90% MCS due to its substantially higher prediction errors, and even the DNN provides only marginal gains relative to the simpler tree-based methods. This suggests that, in the presence of limited training samples and strong observational noise, additional model complexity does not necessarily translate into improved generalisation.

When the study area is expanded to encompass the entire Guangdong–Hong Kong–Macao Greater Bay Area, the situation changes markedly. In the GBA scenario, **ensemble tree models**—and in particular **XGBoost**—demonstrate clear superiority. XGBoost achieves a test-set R^2 of 0.858, accompanied by a reduction in mean absolute error (MAE) to $4.40\,\mu g\,m^{-3}$, indicating both improved explanatory power and tighter error dispersion. Random forest also benefits from the enlarged training set, with an R^2 of 0.831, whereas the neural network models converge to a lower performance plateau around $R^2 = 0.792$. Crucially, the MCS analysis reveals that only XGBoost attains a p-value of 1.0 in the GBA scenario and hence is the sole model retained in the 90% confidence set. Random forest, MLP and DNN are all excluded due to significantly larger squared prediction errors, despite some of them exhibiting nominally high R^2 values. This pattern indicates that, once a sufficiently rich and heterogeneous training dataset is available, gradient-boosted trees are particularly effective at exploiting non-linear interactions between meteorology, precursors and the GEOS-Chem baseline field, whereas both conventional bagging (random forest) and fully connected neural networks appear less efficient in capturing these structures.

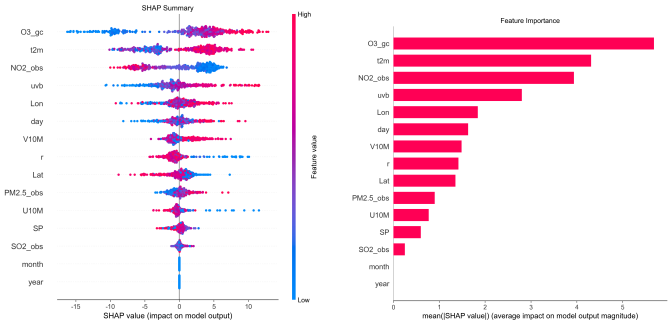
The contrast between the two scenarios underscores the role of spatial information expansion envisioned in the experimental design. In the Hong Kong case, the limited spatial coverage restricts the range of meteorological regimes and emission conditions that the models can learn from, leading to a situation where several algorithms perform similarly and model ranking is largely driven by sampling variability. In the GBA case, the inclusion of multi-city and multi-regime data allows the models to internalise mesoscale meteorological gradients, cross-boundary transport, and regional differences in ozone precursors. The fact that XGBoost is uniquely selected by the MCS as the best-performing specification indicates that its tree-based architecture is particularly well suited to representing these high-dimensional, non-linear interactions under the unified feature space defined in Table I.

Furthermore, comparison with the GEOS-Chem physical–chemical model highlights the potential of the hybrid, data-driven approach. The original GEOS-Chem simulation, when evaluated against station observations over the GBA, yields an R^2 of only 0.29 and exhibits pronounced systematic bias (Fig. 3). In particular, the raw CTM fields tend to underestimate peak ozone events and display substantial residual structure associated with unresolved local processes and emission uncertainties. By contrast, the XGBoost-based bias-correction model, trained on the same observational network but enriched with meteorological and precursor information, substantially reduces these deficiencies. The predictive R^2 is increased to approximately 0.86, corresponding to nearly a threefold gain in explained variance relative to the uncorrected GEOS-Chem output. Residual histograms and observed–predicted scatter plots (see Appendix) confirm that biases at both low and high concentration ranges are strongly mitigated, and that the dispersion of errors is markedly reduced. Taken together, these results demonstrate that the proposed framework combines the large-scale physical consistency of GEOS-Chem with the flexibility of machine learning, yielding a statistically rigorous improvement in ozone forecasting skill over a densely populated, meteorologically complex region.

B. Physical Interpretability Analysis

Having established the relative performance of the candidate algorithms, we next examine whether the best model (**XGBoost**) has genuinely captured the physical essence of

atmospheric processes, rather than merely exploiting statistical artefacts in the training data. To this end, the study employs **SHAP Dependence Plots** and related diagnostics to conduct a detailed deconstruction of the decision-making mechanism, thereby addressing the black-box nature typically associated with machine learning models.



(a) SHAP Summary Plot (b) SHAP Bar Plot

Fig. 4: Global Feature Importance for XGBoost Ozone Prediction

At the global level, SHAP summary and bar plots (Fig. 4a–4b) indicate that the most influential predictors are 2-metre air temperature, GEOS-Chem baseline ozone, ultraviolet radiation, and nitrogen dioxide, followed by surface pressure, humidity and wind components. This ranking is highly consistent with the prior physical understanding summarised in the literature review: temperature and radiation govern photochemical production, NO_2 plays a dual role as both a precursor and a titrant, and the CTM baseline encapsulates the large-scale chemical and transport background. The prominence of the GEOS-Chem ozone field in the SHAP analysis confirms that the hybrid model does not discard the process-based information contained in the CTM, but instead refines it using local meteorological and precursor corrections.

Nonlinear temperature responses and apparent thresholds: SHAP analysis identifies 2-metre air temperature (t_{2m}) as the dominant meteorological factor influencing ozone forecasts. Its SHAP values exhibit a pronounced non-linear increase with rising temperatures (Fig. 5). Particularly above the mean or climatological median temperature, the contribution of t_{2m} to predicted ozone surges sharply, suggesting an effective temperature threshold beyond which photochemical ozone production becomes much more efficient. This behaviour aligns closely with the Arrhenius-type dependence of reaction rates on temperature and with observational studies reporting enhanced ozone episodes during heatwaves. The model therefore appears to have internalised a key physical mechanism rather than learning a purely empirical correlation.

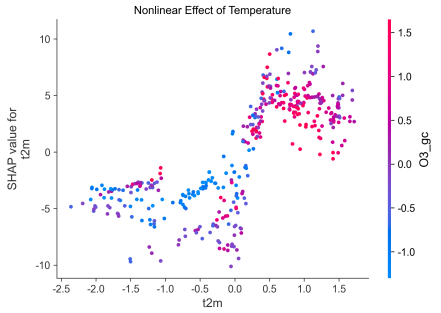


Fig. 5: SHAP Dependence on Temperature (t_{2m})

Synergistic interactions among meteorological factors: The model also successfully captures the strong interaction effect between temperature and ultraviolet radiation (UVB). SHAP interaction plots reveal that, within the high-temperature range, ozone formation potential only peaks when UVB radiation is strong (clear-sky conditions). Conversely, under weak UVB (overcast or heavily polluted conditions), the ozone-enhancing effect of high temperatures is significantly suppressed (Fig. 6). This pattern demonstrates that the model has learned that "high temperature" and "sufficient actinic flux" are jointly necessary conditions for intense photochemical smog formation, in agreement with the conceptual frameworks of ozone chemistry discussed in Section II. Additional interactions involving wind components and humidity indicate that the model also encodes the competing roles of dispersion, mixing and wet scavenging, although these effects are secondary compared with the temperature–radiation synergy.

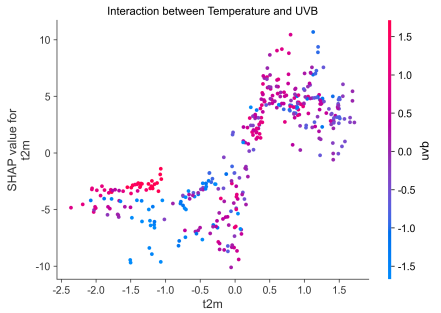


Fig. 6: Interaction between t_{2m} and UVB

Non-linear titration effect of NO_2: For the precursor NO_2, the model exhibits a bidirectional regulatory mechanism that is closely aligned with established atmospheric chemistry. At low to moderate concentrations, NO_2 contributes positively to ozone formation, reflecting its role in the photochemical production cycle. However, at high concentrations (for example in urban centres with strong local emissions), the SHAP values associated with NO_2 abruptly turn strongly negative (Fig. 7). This behaviour mirrors the NO_x titration effect commonly observed in winter and in heavily polluted boundary layers, where excess NO directly consumes ozone:

$$NO + O_3 \rightarrow NO_2 + O_2.$$

The emergence of this characteristic non-linear response from a purely data-driven model provides strong evidence that the trained XGBoost regressor has recovered key mechanistic features of the ozone–NO_x system without these relationships being explicitly hard-coded.

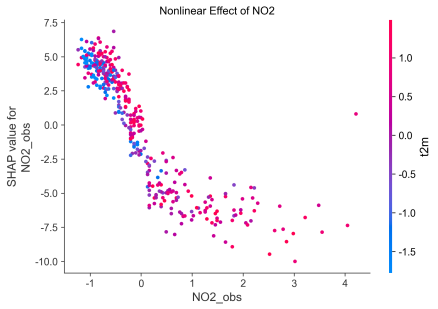


Fig. 7: SHAP Dependence on NO_2 Showing Titration Effect

Finally, spatial predictors (longitude and latitude) and the temporal indicators (year, month, day) exhibit moderate but non-negligible SHAP contributions, reflecting the model's use of these variables to adjust for seasonal cycles, regional background differences and gradually evolving emission patterns. Their relatively lower importance compared with meteorology and precursors is consistent with the expectation that ozone dynamics are primarily driven by short-term physical and chemical conditions, with space–time coordinates acting mainly as structural modifiers.

In summary, the **XGBoost** model not only excels in conventional statistical accuracy metrics but, more importantly, successfully internalises a suite of atmospheric physicochemical mechanisms that are consistent with both GEOS-Chem simulations and observational studies. The SHAP-based diagnostics demonstrate that the downscaling framework behaves in a physically interpretable manner, providing confidence that the improvements over the raw CTM fields arise from scientifically meaningful corrections rather than from overfitting or spurious correlations. This supports the use of the hybrid approach as both a forecasting tool and a diagnostic instrument for regional ozone pollution in the Greater Bay Area.

VI. Conclusion

This study systematically addresses the longstanding limitations of chemical transport models (CTMs) in reproducing near-surface ozone over the Greater Bay Area by developing a high-resolution downscaling framework grounded in machine learning. By integrating multi-source atmospheric information and employing formal statistical evaluation, the proposed framework provides both methodological advances and practical implications for regional air quality forecasting.

First, comprehensive algorithm benchmarking demonstrated the consistent superiority of **ensemble tree models** for structured environmental datasets. Through rigorous **Model Confidence Set (MCS)** inference, we established that **XGBoost** exhibits significantly greater robustness compared with random forests and neural network architectures. Under the regional (GBA-wide) training scenario, XGBoost was the only model retained in the optimal confidence set, achieving a predictive R^2 of 0.86. These results highlight the importance of model architectures capable of capturing non-linear interactions and heterogeneous spatial patterns, which are characteristic of complex atmospheric processes.

Second, the study shows that incorporating spatial diversity is indispensable for improving predictive performance. Experiments contrasting localised Hong Kong training with regionally expanded datasets reveal that models trained on GBA-wide samples successfully internalise cross-city pollutant transport dynamics, mesoscale meteorological gradients, and regional precursor variability. This finding underscores a broader methodological implication: ozone downscaling models benefit substantially from appropriate spatial generalisation, and performance plateaus observed in purely local datasets can be structurally unavoidable.

Finally, the interpretability analysis conducted via **SHAP** provides strong evidence that the downscaling model is not merely fitting statistical associations but is aligned with known atmospheric chemistry. The recovered patterns—temperature-dominated photochemical responses, synergistic meteorological modulations, and the characteristic non-linear NOx titration regime—demonstrate that data-driven learning can recover mechanistic behaviours traditionally expected only in physically based CTMs. This confirms that when properly trained and validated, machine learning models can serve not only as predictive tools but also as diagnostic instruments for understanding process-level dynamics.

Overall, the framework developed in this study delivers a high-precision, computationally efficient ozone forecasting system tailored for the Greater Bay Area. More broadly, it provides a transferable paradigm for integrating **Physics-Guided Machine Learning** into atmospheric research, offering a pathway for bridging physical models and data-driven approaches in next-generation air quality prediction systems.

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APPENDIX

This appendix provides detailed visual evidence supporting the statistical results presented in Section 6.1. It contains scatter plots (Predicted vs. Observed) and residual histograms for all four benchmarked models (Random Forest, XGBoost, MLP, and DNN) across the two experimental scenarios. These visualizations allow for a granular inspection of goodness-of-fit and error distributions.

A.1 Hong Kong Scenario: Model Comparison

Figures 8–11 illustrate performance in the **data-limited Hong Kong scenario**.

- **Observation:** Scatter plots show wider dispersion around the 1:1 line than the regional scenario, confirming the challenge of predicting ozone using solely local data without spatial context.
- **Model Comparison:** While tree-based models (RF, XGBoost) maintain linear trends, neural networks (MLP, DNN; Figs. 10, 11) exhibit more scatter at high concentrations. This supports MCS findings that neural networks are less robust in small-sample regimes.

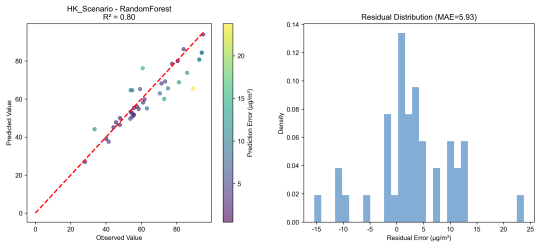


Fig. 8: HK Scenario — Random Forest: Scatter plot (left) and residual distribution (right).

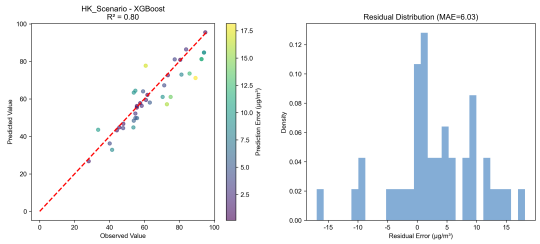


Fig. 9: HK Scenario — XGBoost: Scatter plot (left) and residual distribution (right).

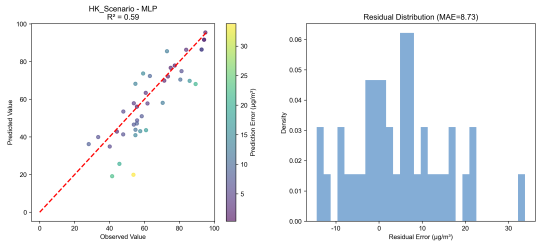


Fig. 10: HK Scenario — MLP: Scatter plot (left) and residual distribution (right).

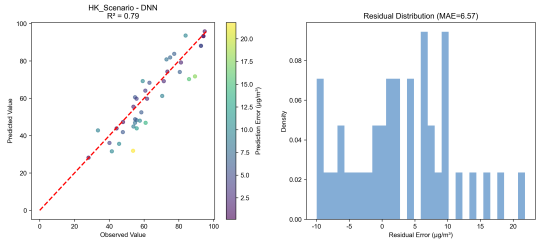


Fig. 11: HK Scenario — DNN: Scatter plot (left) and residual distribution (right).

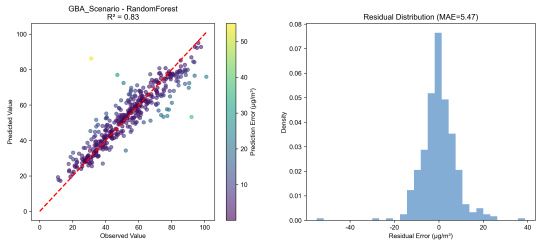


Fig. 12: GBA Scenario — Random Forest: Scatter plot (left) and residual distribution (right).

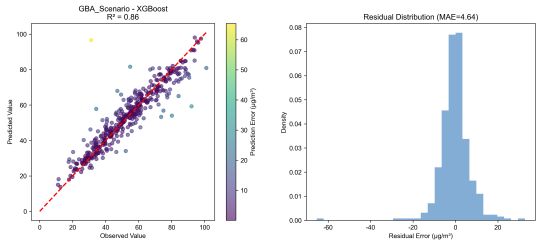


Fig. 13: GBA Scenario — XGBoost: Scatter plot (left) and residual distribution (right).

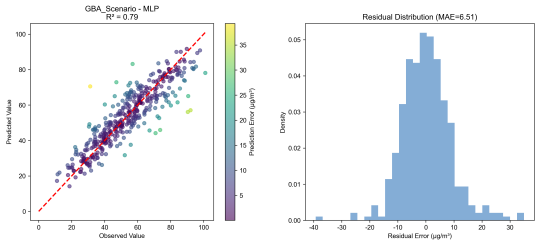


Fig. 14: GBA Scenario — MLP: Scatter plot (left) and residual distribution (right).

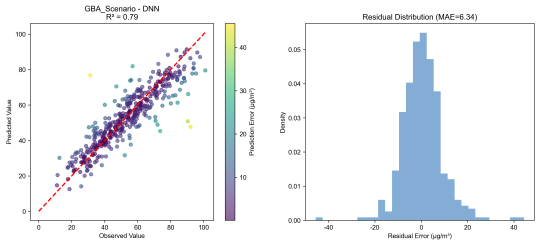


Fig. 15: GBA Scenario — DNN: Scatter plot (left) and residual distribution (right).

A.2 Greater Bay Area Scenario: Model Comparison

Figures 12 through 15 present the results for the **data-rich Greater Bay Area Regional Scenario**.

- **Observation:** A significant improvement in prediction accuracy is visually evident across all models. The data points cluster much more tightly along the red dashed line, and the residual histograms are narrower and more peaked around zero. This confirms the study's core finding that expanding the spatial scope allows models to learn regional transport patterns, thereby reducing prediction error.
- **Champion Model:** Notably, Figure 13 (XGBoost) demonstrates the highest density of points along the diagonal and the most symmetrical residual distribution. This visually supports the statistical conclusion that XGBoost is the superior model for this task, effectively minimizing systematic bias compared to other algorithms.

AUTHOR CONTRIBUTIONS

The authors contributed to this work as follows (in no particular order):

Wentao Wang was responsible for the data collection, modeling framework;

Lifeng Han was responsible for the code implementation and image painting;

Chenxi Pan was responsible for the manuscript writing and theoretical analysis.

All authors contributed equally to this study.